

HCSM-45

Graviton as Self Paired Curvature Mode

The Bilinear $h = \omega \otimes \omega$ and the $54B_1 \oplus 54B_2$ Mode Count The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model realizes one Standard Model generation on the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus (HCSM-04), and its gauge structure on the full 56-dimensional Hodge sector $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$ (HCSM-02, HCSM-06). This paper identifies the graviton as the *self-paired curvature mode* of the substrate: the symmetric bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, projected onto the $B_1 \oplus B_2$ isotypic sector of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$. We prove: (i) the D_4 isotypic decomposition of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ is

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E,$$

of total dimension $60 + 46 + 54 + 54 + 192 = 406 = 28 \cdot 29/2$; (ii) the graviton sector is the $B_1 \oplus B_2$ isotypic piece, comprising $54B_1 \oplus 54B_2 = 108$ modes; (iii) the graviton polarization space is $V = B_1 \oplus B_2$, of dimension 2, with D_4 character $(2, 2, -2, 0, 0)$ on the five conjugacy classes; (iv) the B_1 component corresponds to the $h_{xx} - h_{yy}$ polarization and the B_2 component to the h_{xy} polarization. The result is HCSM-native: it uses only the Hodge complex of HCSM-02, the D_4 representation theory of HCSM-05 and HCSM-06, and the standard tensor square character formula. No imported gravitational machinery appears. All claims are verified at machine precision.

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1 Introduction

1.1 The graviton problem

The Hodge Complex Standard Model (HCSM) derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ (HCSM-06 [6]) and identifies the physical hypercharge functional on the sector-coarse space (HCSM-07 [?]). The full 56-dimensional Hodge sector at $\lambda = 4$ carries the D_4 content

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E \quad (1)$$

(HCSM-02 [3], HCSM-05 [5]).

The graviton in general relativity is a massless spin-2 field $h_{\mu\nu}$, identified with the metric fluctuation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. In the HCSM framework, the natural home of the graviton is the *symmetric tensor square* $\text{Sym}^2(\Omega^1)$ of the 1-form sector: this is the discrete analogue of the space of symmetric 2-tensors on the torus.

This paper establishes the graviton as the self paired curvature mode of the substrate: the bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, symmetrized and projected onto the $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.

1.2 The one-line summary

The graviton is the symmetric bilinear $h = \omega \otimes \omega$ in the substrate 1-form ω , projected onto the $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$. The sector has $54B_1 \oplus 54B_2 = 108$ modes, with polarization space $V = B_1 \oplus B_2$ of dimension 2.

1.3 Main results

- (1) **Symmetric-square decomposition** (Theorem 3.4). The D_4 isotypic decomposition of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ is

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E. \quad (2)$$

The total dimension is $406 = 28 \cdot 29/2$, matching $\dim \text{Sym}^2(\mathbb{C}^{28})$.

- (2) **Graviton sector** (Corollary 4.2). The $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ has dimension 108 and decomposes as

$$V_{\text{grav}} = 54B_1 \oplus 54B_2. \quad (3)$$

- (3) **Polarization space** (Theorem 5.1). The graviton polarization space is $V = B_1 \oplus B_2$, of dimension 2, with D_4 character $(2, 2, -2, 0, 0)$ on the five conjugacy classes of D_4 .
- (4) **Physical polarization** (Corollary 5.3). The B_1 component corresponds to the $h_{xx} - h_{yy}$ polarization and the B_2 component to the h_{xy} polarization.
- (5) **Self-paired bilinear** (Theorem 6.2). The graviton is the symmetrized self-paired bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, restricted to the $B_1 \oplus B_2$ isotypic sector.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex; the $\Omega^1|_{\lambda=4}$ decomposition.
- HCSM-04 [4]: the 14-mode multiplet.
- HCSM-05 [5]: orbit decomposition; the D_4 character table.
- HCSM-06 [6]: sector-coarse correspondence; the D_4 tensor-product rules.
- HCSM-46 [7]: graviton exact masslessness (downstream).
- HCSM-47 [8]: graviton self-coupling from D_4 (downstream).
- HCSM-48 [9]: Faddeev-Popov G_N normalization (downstream).

1.5 Organization

Section 2 states the Hodge complex, the $\Omega^1|_{\lambda=4}$ decomposition, and the D_4 character table. Section 3 proves the symmetric-square decomposition. Section 4 identifies the graviton sector. Section 5 establishes the polarization space. Section 6 defines the self-paired bilinear. Section 7 summarizes inputs and derived results. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup

2.1 The Hodge complex at $\lambda = 4$

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01 [2]). The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (4)$$

of total dimension 256 (HCSM-02 [3]). The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree.

At the self-paired eigenvalue $\lambda = 4$, the Hodge complex decomposes under D_4 as (HCSM-02 [3], HCSM-05 [5])

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (5)$$

$$\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (6)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (7)$$

The sector of interest in this paper is

$$\Omega^1|_{\lambda=4} = \Omega^1|_{\lambda=4}, \quad \dim \Omega^1|_{\lambda=4} = 3 + 3 + 3 + 3 + 16 = 28. \quad (8)$$

2.2 The D_4 character table

The D_4 irreps are A_1, A_2, B_1, B_2, E (four 1-dimensional, one 2-dimensional). The conjugacy classes are $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$ with class sizes $(1, 1, 2, 2, 2)$:

irrep ρ	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

(Columns are labeled by a representative of the class.)

2.3 The D_4 tensor-product rules

The tensor product of D_4 irreps follows the standard rules. The relevant ones for this paper are:

$$A_1 \otimes \rho = \rho \quad \text{for any irrep } \rho, \quad (9)$$

$$A_2 \otimes A_2 = A_1, \quad (10)$$

$$A_2 \otimes B_1 = B_2, \quad A_2 \otimes B_2 = B_1, \quad (11)$$

$$A_2 \otimes E = E, \quad (12)$$

$$B_1 \otimes B_1 = A_1, \quad B_2 \otimes B_2 = A_1, \quad (13)$$

$$B_1 \otimes B_2 = A_2, \quad (14)$$

$$B_i \otimes E = E \quad \text{for } i = 1, 2, \quad (15)$$

$$E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2. \quad (16)$$

In particular, the symmetric and antisymmetric parts of $E \otimes E$ are

$$\text{Sym}^2(E) = A_1 \oplus B_1, \quad \Lambda^2(E) = A_2 \oplus B_2. \quad (17)$$

2.4 The symmetric-square character formula

For a finite group G acting on a finite-dimensional representation V , the character of $\text{Sym}^2(V)$ on a group element g is

$$\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2} [\chi_V(g)^2 + \chi_V(g^2)]. \quad (18)$$

The multiplicities of the irreps in $\text{Sym}^2(V)$ follow from the orthogonality relation

$$m_\rho = \frac{1}{|G|} \sum_C |C| \chi_{\text{Sym}^2(V)}(C) \chi_\rho(C), \quad (19)$$

where the sum runs over conjugacy classes C and $|C|$ is the class size.

3 The symmetric-square decomposition

3.1 Sym^2 of the individual isotypic pieces

Let $\Omega^1|_{\lambda=4} = \bigoplus_i V_i$ with $V_i = m_i \rho_i$ the isotypic pieces. Then

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = \bigoplus_i \text{Sym}^2(V_i) \oplus \bigoplus_{i < j} (V_i \otimes V_j). \quad (20)$$

We compute each piece for $\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E$.

Lemma 3.1 (Sym^2 of the 1-dimensional pieces). *For $V_i = 3\rho_i$ with $\rho_i \in \{A_1, A_2, B_1, B_2\}$,*

$$\text{Sym}^2(3\rho_i) = 6A_1. \quad (21)$$

Proof. Take $\rho_i = B_1$ (the other cases are identical). Then $V = 3B_1$ is a 3-dimensional space with D_4 action via the character χ_{B_1} on each basis vector. So $\chi_V(g) = 3\chi_{B_1}(g)$, giving character $(3, 3, -3, 3, -3)$ on the five classes.

Using (18):

$$\chi_{\text{Sym}^2(V)}(e) = (3^2 + 3)/2 = 6, \quad (22)$$

$$\chi_{\text{Sym}^2(V)}(r^2) = (3^2 + 3)/2 = 6, \quad (23)$$

$$\chi_{\text{Sym}^2(V)}(r) = ((-3)^2 + 3)/2 = 6, \quad (24)$$

$$\chi_{\text{Sym}^2(V)}(s) = (3^2 + 3)/2 = 6, \quad (25)$$

$$\chi_{\text{Sym}^2(V)}(rs) = ((-3)^2 + 3)/2 = 6. \quad (26)$$

Hence $\chi_{\text{Sym}^2(V)} = (6, 6, 6, 6, 6)$, which by (19) decomposes as $6A_1$. Dimension check: $6 = \dim \text{Sym}^2(3\text{-dim}) = 3 \cdot 4/2 = 6$. $\square$

Lemma 3.2 (Sym^2 of $8E$).

$$\text{Sym}^2(8E) = 36A_1 \oplus 28A_2 \oplus 36B_1 \oplus 36B_2. \quad (27)$$

Proof. $V = 8E$ has dimension 16 and character $\chi_V = (16, -16, 0, 0, 0)$. Using (18):

$$\chi_{\text{Sym}^2(V)}(e) = (16^2 + 16)/2 = 136, \quad (28)$$

$$\chi_{\text{Sym}^2(V)}(r^2) = ((-16)^2 + 16)/2 = 136, \quad (29)$$

$$\chi_{\text{Sym}^2(V)}(r) = (0^2 + (-16))/2 = -8, \quad (30)$$

$$\chi_{\text{Sym}^2(V)}(s) = (0^2 + 16)/2 = 8, \quad (31)$$

$$\chi_{\text{Sym}^2(V)}(rs) = (0^2 + 16)/2 = 8. \quad (32)$$

So $\chi_{\text{Sym}^2(V)} = (136, 136, -8, 8, 8)$.

Applying (19):

$$m_{A_1} = \frac{1}{8}[136 + 136 + 2(-8) + 2(8) + 2(8)] = 36, \quad (33)$$

$$m_{A_2} = \frac{1}{8}[136 + 136 + 2(-8) + 2(-8) + 2(-8)] = 28, \quad (34)$$

$$m_{B_1} = \frac{1}{8}[136 + 136 + 2(8) + 2(8) + 2(-8)] = 36, \quad (35)$$

$$m_{B_2} = \frac{1}{8}[136 + 136 + 2(8) + 2(-8) + 2(8)] = 36, \quad (36)$$

$$m_E = \frac{1}{8}[136 \cdot 2 + 136 \cdot (-2)] = 0. \quad (37)$$

Dimension check: $36 + 28 + 36 + 36 + 0 = 136 = \dim \text{Sym}^2(16\text{-dim})$. $\square$

Lemma 3.3 (Cross terms). *The pairwise cross terms of $\Omega^1|_{\lambda=4}$ are*

$$3A_1 \otimes 3A_2 = 9A_2, \quad (38)$$

$$3A_1 \otimes 3B_1 = 9B_1, \quad (39)$$

$$3A_1 \otimes 3B_2 = 9B_2, \quad (40)$$

$$3A_1 \otimes 8E = 24E, \quad (41)$$

$$3A_2 \otimes 3B_1 = 9B_2, \quad (42)$$

$$3A_2 \otimes 3B_2 = 9B_1, \quad (43)$$

$$3A_2 \otimes 8E = 24E, \quad (44)$$

$$3B_1 \otimes 3B_2 = 9A_2, \quad (45)$$

$$3B_1 \otimes 8E = 24E, \quad (46)$$

$$3B_2 \otimes 8E = 24E. \quad (47)$$

Proof. Direct application of the tensor-product rules of Section 2. For example, $3A_1 \otimes 3A_2 = 9 \cdot (A_1 \otimes A_2) = 9A_2$, and $3A_2 \otimes 3B_1 = 9 \cdot (A_2 \otimes B_1) = 9B_2$. $\square$

3.2 The full decomposition

Theorem 3.4 (Symmetric-square decomposition). *The D_4 isotypic decomposition of $\text{Sym}^2(\Omega^1|_{\lambda=4})$ is*

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E. \quad (48)$$

The total dimension is 406, matching $\dim \text{Sym}^2(\mathbb{C}^{28}) = 28 \cdot 29/2$.

Proof. We sum the contributions from the individual pieces (Lemma 3.1, Lemma 3.2) and the cross terms (Lemma 3.3).

Sym² of the individual isotypic pieces:

$$\text{Sym}^2(3A_1) = 6A_1, \quad (49)$$

$$\text{Sym}^2(3A_2) = 6A_1, \quad (50)$$

$$\text{Sym}^2(3B_1) = 6A_1, \quad (51)$$

$$\text{Sym}^2(3B_2) = 6A_1, \quad (52)$$

$$\text{Sym}^2(8E) = 36A_1 \oplus 28A_2 \oplus 36B_1 \oplus 36B_2. \quad (53)$$

Sum: $24A_1 \oplus 0A_2 \oplus 0B_1 \oplus 0B_2 \oplus 0E$ plus $36A_1 \oplus 28A_2 \oplus 36B_1 \oplus 36B_2 = 60A_1 \oplus 28A_2 \oplus 36B_1 \oplus 36B_2$.

Cross terms: From Lemma 3.3, the cross terms contribute

$$A_2 : 9(\text{from } 3A_1 \otimes 3A_2) + 9(\text{from } 3B_1 \otimes 3B_2) = 18, \quad (54)$$

$$B_1 : 9(\text{from } 3A_1 \otimes 3B_1) + 9(\text{from } 3A_2 \otimes 3B_2) = 18, \quad (55)$$

$$B_2 : 9(\text{from } 3A_1 \otimes 3B_2) + 9(\text{from } 3A_2 \otimes 3B_1) = 18, \quad (56)$$

$$E : 24 \cdot 4 = 96. \quad (57)$$

Total:

$$A_1 : 60, \quad (58)$$

$$A_2 : 28 + 18 = 46, \quad (59)$$

$$B_1 : 36 + 18 = 54, \quad (60)$$

$$B_2 : 36 + 18 = 54, \quad (61)$$

$$E : 96. \quad (62)$$

Dimension check: $60 + 46 + 54 + 54 + 2 \cdot 96 = 406 = 28 \cdot 29/2$. $\square$

Remark 3.5 (The 54 in context). The 54 appears as $36 + 18$: 36 from $\text{Sym}^2(8E)$ and 18 from cross terms of the 1-dimensional pieces. This is the source of the $54B_1 \oplus 54B_2$ count that the handoff records. The number $54 = 36 + 9 + 9$ has no obviously simpler algebraic form in framework constants; it is the exact result of the D_4 representation-theoretic computation.

4 The graviton sector

Definition 4.1 (Graviton sector). The *graviton sector* of the Hodge complex is the $B_1 \oplus B_2$ isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.

Corollary 4.2 (Graviton mode count). *The graviton sector has dimension 108 and decomposes as*

$$V_{\text{grav}} = 54B_1 \oplus 54B_2. \quad (63)$$

Proof. Immediate from Theorem 3.4: the B_1 and B_2 multiplicities are both 54. Dimension: $54+54 = 108$. $\square$

Remark 4.3 (Why $B_1 \oplus B_2$?). The B_1 and B_2 irreps are the two 1-dimensional nontrivial representations of D_4 that transform as the two physical polarizations of the graviton in the discrete setting: $h_{xx} - h_{yy}$ (B_1) and h_{xy} (B_2). The A_1 and A_2 pieces correspond to the trace and the antisymmetric part of the tensor; the E piece corresponds to vector content that is not the graviton.

5 Polarization space

Theorem 5.1 (Polarization space). *The graviton polarization space is*

$$V = B_1 \oplus B_2, \quad (64)$$

of dimension 2, with D_4 character

$$\chi_V = (2, 2, -2, 0, 0) \quad (65)$$

on the five conjugacy classes (e, r^2, r, s, rs) .

Proof. By definition, the graviton sector is the $B_1 \oplus B_2$ isotypic piece. As a D_4 representation, it is $V = B_1 \oplus B_2$ (the polarization space), tensored with the multiplicity space of dimension 54 (in each isotypic component).

The character of $V = B_1 \oplus B_2$ is

$$\chi_V(e) = 1 + 1 = 2, \quad (66)$$

$$\chi_V(r^2) = 1 + 1 = 2, \quad (67)$$

$$\chi_V(r) = (-1) + (-1) = -2, \quad (68)$$

$$\chi_V(s) = 1 + (-1) = 0, \quad (69)$$

$$\chi_V(rs) = (-1) + 1 = 0. \quad (70)$$

Hence $\chi_V = (2, 2, -2, 0, 0)$. □

Remark 5.2 (Physical identification). The two components of V correspond to the two physical polarizations of the graviton:

- B_1 : the $h_{xx} - h_{yy}$ polarization.
- B_2 : the h_{xy} polarization.

These are the two transverse-traceless polarizations of a massless spin-2 field in $d = 4$ dimensions.

Corollary 5.3 (Physical polarization). *The B_1 component of the graviton polarization space corresponds to the $h_{xx} - h_{yy}$ polarization and the B_2 component to the h_{xy} polarization.*

Proof. By the standard identification of the B_1 and B_2 irreps with the two independent components of a symmetric traceless 2×2 tensor in two dimensions (HCSM-02 [3]). □

6 The self-paired bilinear

Definition 6.1 (Self-paired bilinear). The *self-paired bilinear* of the substrate 1-form $\omega \in \Omega^1$ is

$$h = \omega \otimes \omega, \quad (71)$$

where the same 1-form ω appears on both sides of the tensor product. The bilinear is *self-paired* in the sense that it is built from ω paired with itself; it contrasts with a generic bilinear $\omega_1 \otimes \omega_2$ with $\omega_1 \neq \omega_2$.

Theorem 6.2 (Graviton as self-paired bilinear). *The graviton is the symmetrization of the self-paired bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, restricted to the $\lambda = 4$ sector and projected onto the $B_1 \oplus B_2$ isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.*

Proof. The symmetrization of $h = \omega \otimes \omega$ gives an element of $\text{Sym}^2(\Omega^1)$. At $\lambda = 4$, this restricts to $\text{Sym}^2(\Omega^1|_{\lambda=4})$. By Theorem 3.4, $\text{Sym}^2(\Omega^1|_{\lambda=4})$ decomposes as $60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E$. The projection onto the $B_1 \oplus B_2$ piece is the graviton sector $V_{\text{grav}} = 54B_1 \oplus 54B_2$ (Corollary 4.2).

The identification of this projected bilinear with the physical graviton is established by the following properties, proved in companion papers:

- The polarization space $V = B_1 \oplus B_2$ matches the two physical graviton polarizations (Theorem 5.1, Corollary 5.3).
- The graviton is exactly massless by D_4 Schur orthogonality (HCSM-46 [7]).
- The graviton self-coupling is determined by the D_4 structure of the bilinear (HCSM-47 [8]).

□

Remark 6.3 (Self-paired vs. generic bilinear). A generic bilinear $\omega_1 \otimes \omega_2$ with $\omega_1 \neq \omega_2$ would live in the full tensor square $\Omega^1 \otimes \Omega^1$, of dimension 784, and would have a different D_4 decomposition. The self-paired bilinear $h = \omega \otimes \omega$ is the natural discrete analogue of the metric fluctuation $h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu}$, which is also a self-paired object built from the metric itself.

Remark 6.4 (The term "curvature mode"). The word "curvature" in the paper's title refers to the fact that the bilinear h is the field-theoretic analogue of a curvature fluctuation: it is the symmetrized product of two copies of the substrate 1-form ω , which plays the role of the vielbein in the discrete framework. The curvature interpretation is developed further in HCSM-47 (self-coupling) and HCSM-50 (composite-operator effective action).

7 Inputs and derived

7.1 Derived

The following results are proved in this paper:

1. The D_4 isotypic decomposition of $\text{Sym}^2(\Omega^1|_{\lambda=4})$: $60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E$ (Theorem 3.4).
2. The graviton sector $V_{\text{grav}} = 54B_1 \oplus 54B_2$, of dimension 108 (Corollary 4.2).
3. The polarization space $V = B_1 \oplus B_2$, of dimension 2, with character $(2, 2, -2, 0, 0)$ (Theorem 5.1).
4. The physical identification $B_1 \leftrightarrow h_{xx} - h_{yy}$, $B_2 \leftrightarrow h_{xy}$ (Corollary 5.3).
5. The graviton as the self-paired bilinear $h = \omega \otimes \omega$ (Theorem 6.2).

7.2 Inputs

1. The 8×8 torus substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and its Hodge complex (HCSM-01 [2], HCSM-02 [3]).
2. The D_4 isotypic decomposition $\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E$ (HCSM-02 [3], HCSM-05 [5]).
3. The D_4 character table and tensor-product rules (HCSM-05 [5], HCSM-06 [6]).
4. The symmetric-square character formula $\chi_{\text{Sym}^2(V)}(g) = (\chi_V(g)^2 + \chi_V(g^2))/2$ and the multiplicity formula (19). These are standard representation-theoretic facts.

7.3 What is not claimed

We do not claim to prove:

- That the graviton is exactly massless. This is the subject of HCSM-46 [7].
- The graviton self-coupling structure. This is the subject of HCSM-47 [8].
- The Faddeev-Popov normalization or the Newton constant. These are the subjects of HCSM-48 [9] and HCSM-49 [10].
- The composite-operator effective action. This is the subject of HCSM-50 [11].

This paper establishes only the mode count and the polarization structure of the graviton as a specific D_4 isotypic piece of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.

8 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision, and are exact in rational arithmetic. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-45_verification.py`. It constructs the D_4 character table, computes Sym^2 of each isotypic piece of $\Omega^1|_{\lambda=4}$, and verifies:

1. The character of each isotypic piece.
2. The character of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.
3. The multiplicities $(60, 46, 54, 54, 96)$ of (A_1, A_2, B_1, B_2, E) .
4. The dimension check $60 + 46 + 54 + 54 + 2 \cdot 96 = 406$.
5. The graviton polarization character $(2, 2, -2, 0, 0)$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

9 Conclusion

This paper has established the graviton as the self-paired curvature mode of the Hodge complex on the 8×8 torus.

The graviton is the symmetrized self-paired bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, restricted to the $\lambda = 4$ sector and projected onto the $B_1 \oplus B_2$ isotypic piece of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$.

The symmetric square $\text{Sym}^2(\Omega^1|_{\lambda=4})$ has D_4 isotypic decomposition

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E, \quad (72)$$

of total dimension 406. The graviton sector is

$$V_{\text{grav}} = 54B_1 \oplus 54B_2, \quad \dim V_{\text{grav}} = 108. \quad (73)$$

The polarization space is $V = B_1 \oplus B_2$, of dimension 2, with character $(2, 2, -2, 0, 0)$ on the five conjugacy classes of D_4 . The B_1 component corresponds to the $h_{xx} - h_{yy}$ polarization and the B_2 component to the h_{xy} polarization.

The result is HCSM-native: it uses only the Hodge complex of HCSM-02, the D_4 representation theory of HCSM-05 and HCSM-06, and the standard tensor-square character formula. It is the structural foundation for the downstream papers HCSM-46 (masslessness), HCSM-47 (self-coupling), HCSM-48 (Faddeev-Popov normalization), HCSM-49 (Newton constant 8π factor), and HCSM-50 (composite-operator effective action).

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A Web Crawler Metadata

A.1 Dublin Core

DC.title: HCSM-45: Graviton as Self-Paired Curvature Mode
 DC.creator: Preschutti, Stanley
 DC.creator.orcid: 0009-0004-5445-1744
 DC.date.issued: 2026-09
 DC.type: Text / Preprint / White Paper
 DC.format: application/pdf; text/x-latex

DC.language: en
 DC.publisher: Entropia Research Institute
 DC.rights: All rights reserved
 DC.subject: Hodge Complex Standard Model; HCSM; graviton; self-paired bilinear; Sym^2 of Ω^1 ; $54 B_1 + 54 B_2 = 108$ modes; D4 representation theory; polarization space; 8×8 torus
 DC.description: The Hodge Complex Standard Model identifies the graviton as the self-paired curvature mode of the Hodge complex on the periodic 8×8 torus. The graviton is the symmetrized bilinear $h = \omega$ tensor ω in the substrate 1-form ω , projected onto the $B_1 + B_2$ isotypic sector of $\text{Sym}^2(\Omega^1 | \lambda=4)$. The symmetric square decomposes as $60 A_1 + 46 A_2 + 54 B_1 + 54 B_2 + 96 E$; the graviton sector has $54 B_1 + 54 B_2 = 108$ modes. The polarization space is $V = B_1 + B_2$, of dimension 2, with character $(2, 2, -2, 0, 0)$.
 DC.identifier: HCSM-45-2026-09

A.2 Schema.org JSON-LD

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    " $\text{Sym}^2$  of  $\Omega^1$ ",
    " $54 B_1 + 54 B_2$ ",
    "D4 representation theory",
    "polarization space"
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  "isPartOf": {
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    "name": "HCSM White Paper Series"
  }
}

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A.3 BibTeX for Automated Indexing

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  title = {{HCSM-45}: Graviton as Self-Paired Curvature Mode},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, graviton, self-paired
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    polarization space},
  abstract = {The graviton is the symmetrized self-paired bilinear
     $h = \omega$  tensor  $\omega$  in the substrate 1-form  $\omega$ , restricted
    to the  $\lambda = 4$  sector and projected onto the B1 + B2 isotypic
    piece of  $\text{Sym}^2(\Omega^1|_{\lambda=4})$ . The symmetric square decomposes
    as  $60 A1 + 46 A2 + 54 B1 + 54 B2 + 96 E$ ; the graviton sector has
     $54 B1 + 54 B2 = 108$  modes. The polarization space is  $V = B1 + B2$ ,
    of dimension 2, with character  $(2, 2, -2, 0, 0)$ .}
}

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